Responsible Use of Large Language Models for Information Search

Introduction to Computer Science – TA Lecture Week 5

Lecturer: Kangbo Lyu

14th Oct. 2026

Goal of this Lecture



Use Web Interface to Complete Common Learning Tasks



Understand Large Language Models (LLMs)



Identify Common Hallucination Risks in LLM Outputs



Ethically Use LLM Assistance in All Academic Work

Outline

1. Web Interface Operations and Learning Task Prompt Examples
2. Hallucination Risks and Actionable Verification Strategies
3. Academic Integrity Rules and Responsible Usage Guidelines
4. LLM-Assisted Paper Reading and Textbook Understanding

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Part 1: Web Interface and Prompting



Web Interface Basic Access: DeepSeek as an Example

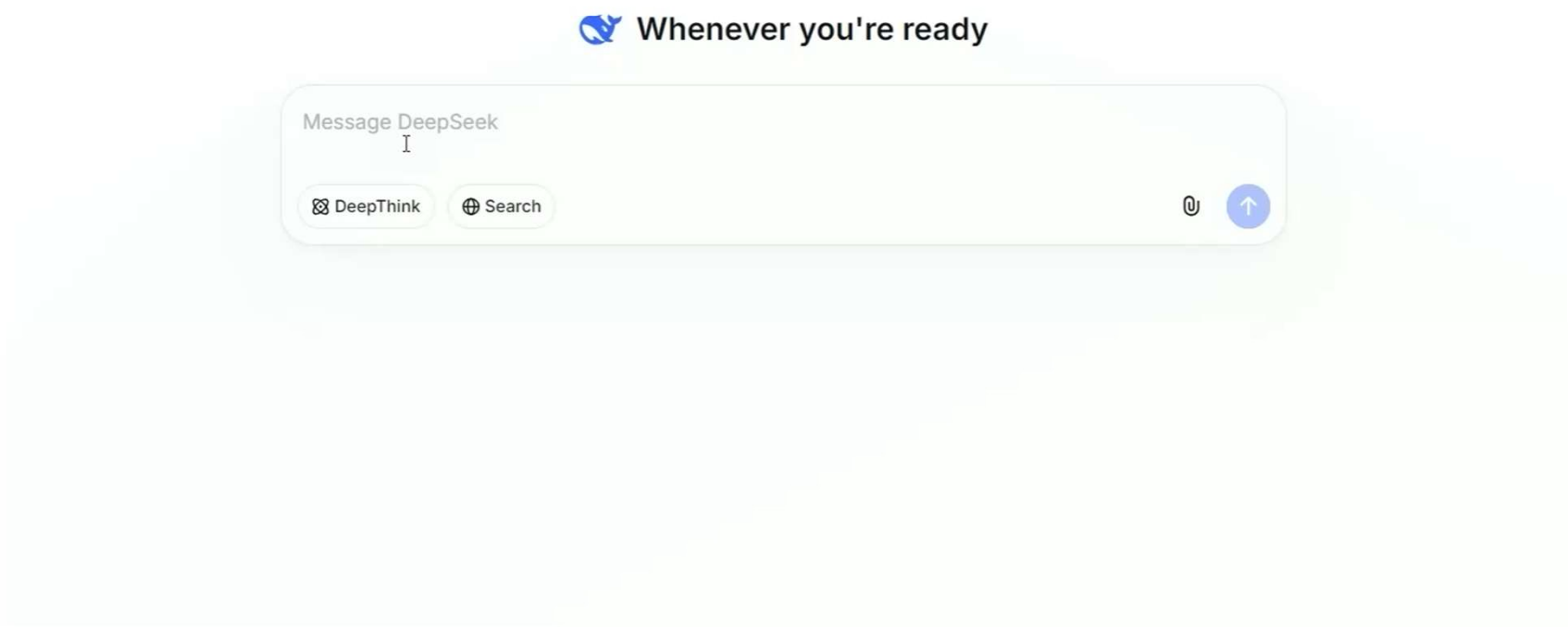


Key Interface Functions for Learning Tasks

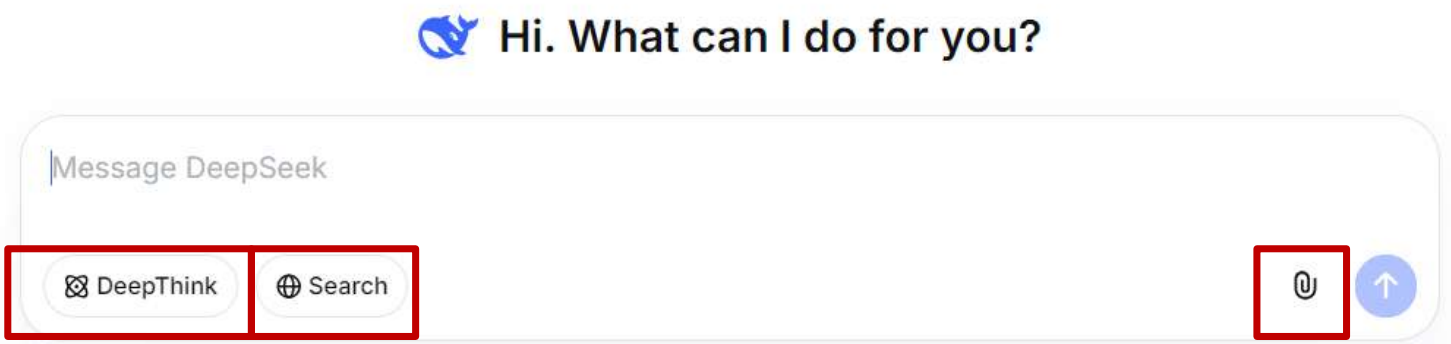


Specific Prompt Examples of Computer Science Learning

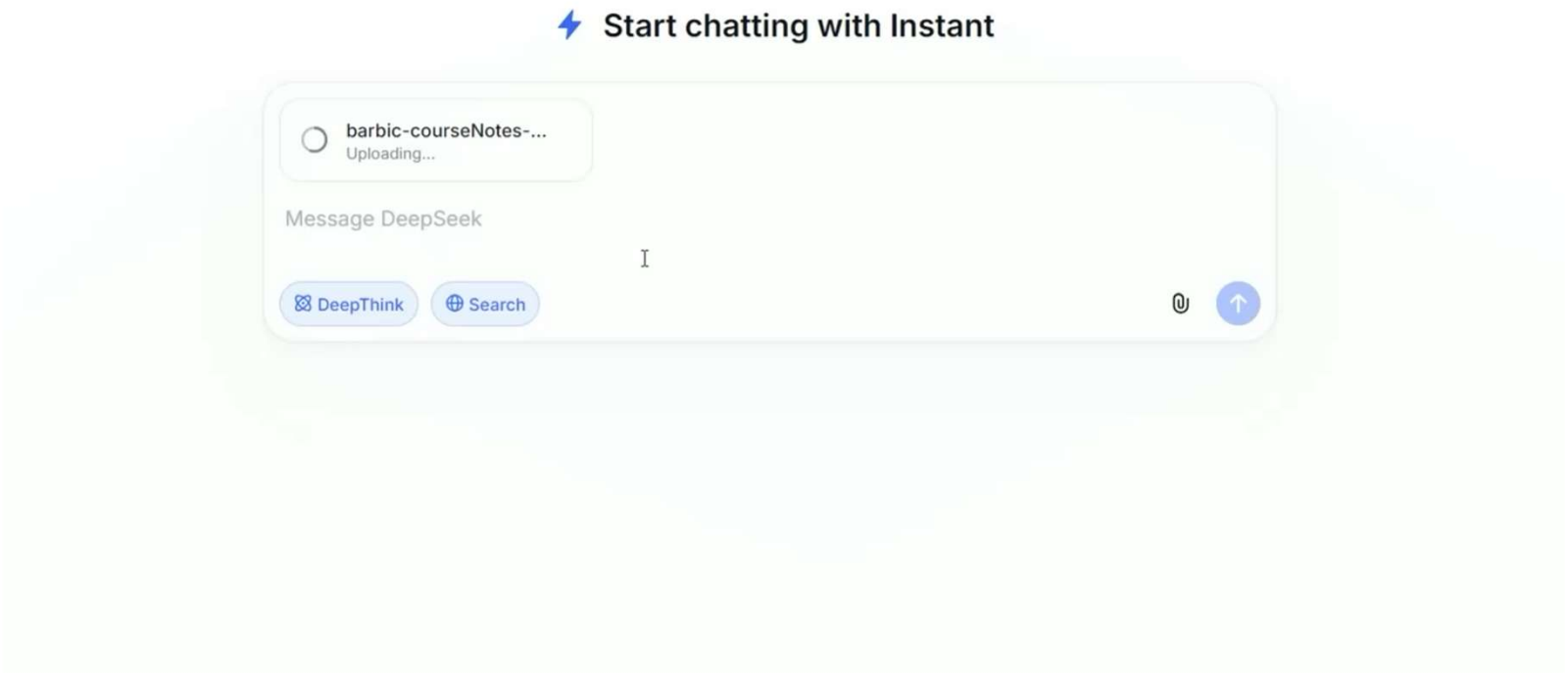
Web Interface: DeepSeek as an Example



Key Interface Functions



Example Usage of Prompting



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Part 2: Hallucination Risks

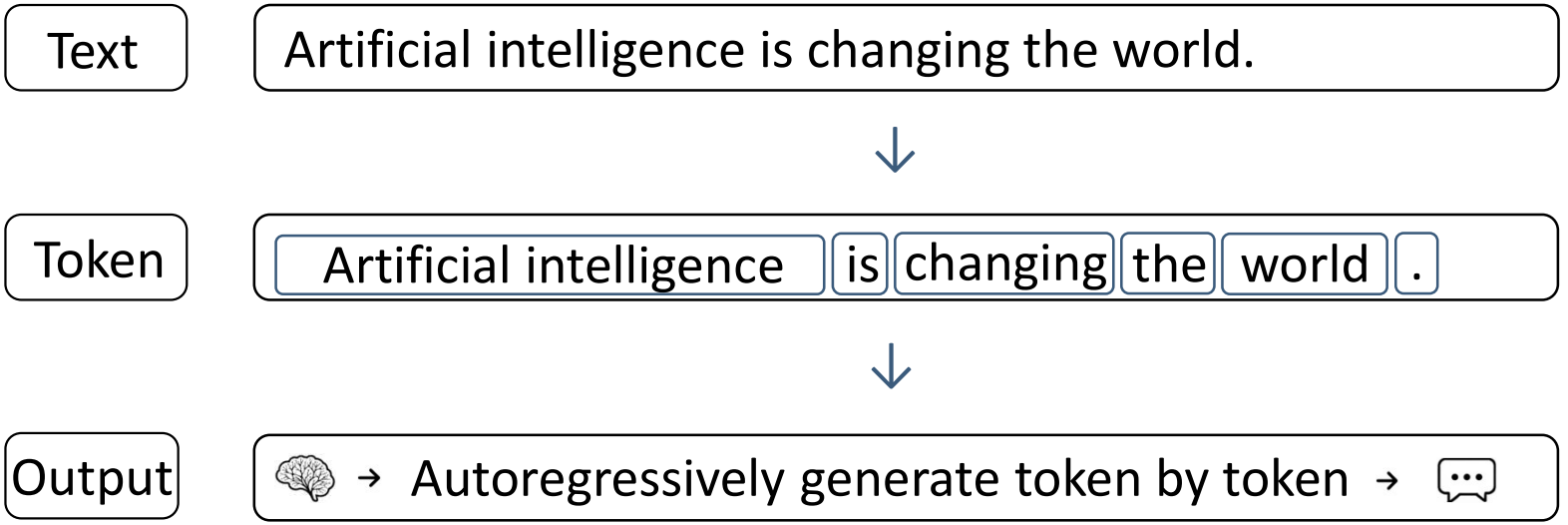
Formal Definition of LLM Hallucinations

Cases of Negative Consequences Caused by Hallucinations

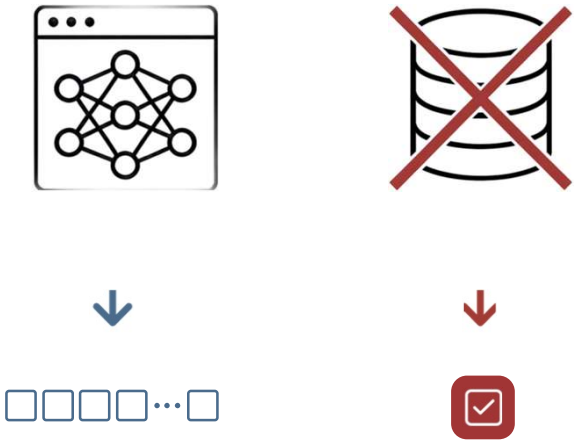
Actionable Checklist for Verifying LLM Outputs



Text Processing in LLMs



Statistical Pattern Learner, not Database



LLMs are trained on large-scale public text

Learn statistical patterns of human language

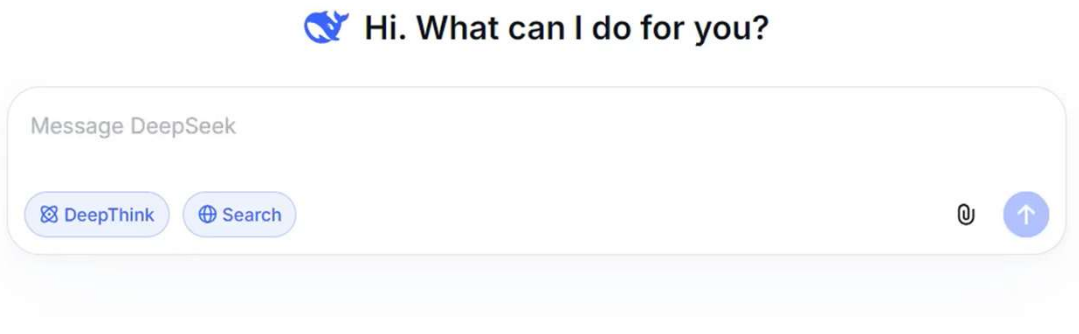
Predict the statistically probable next token

Generate fluent text, but not verified fact

Hallucination: Fluent but Incorrect

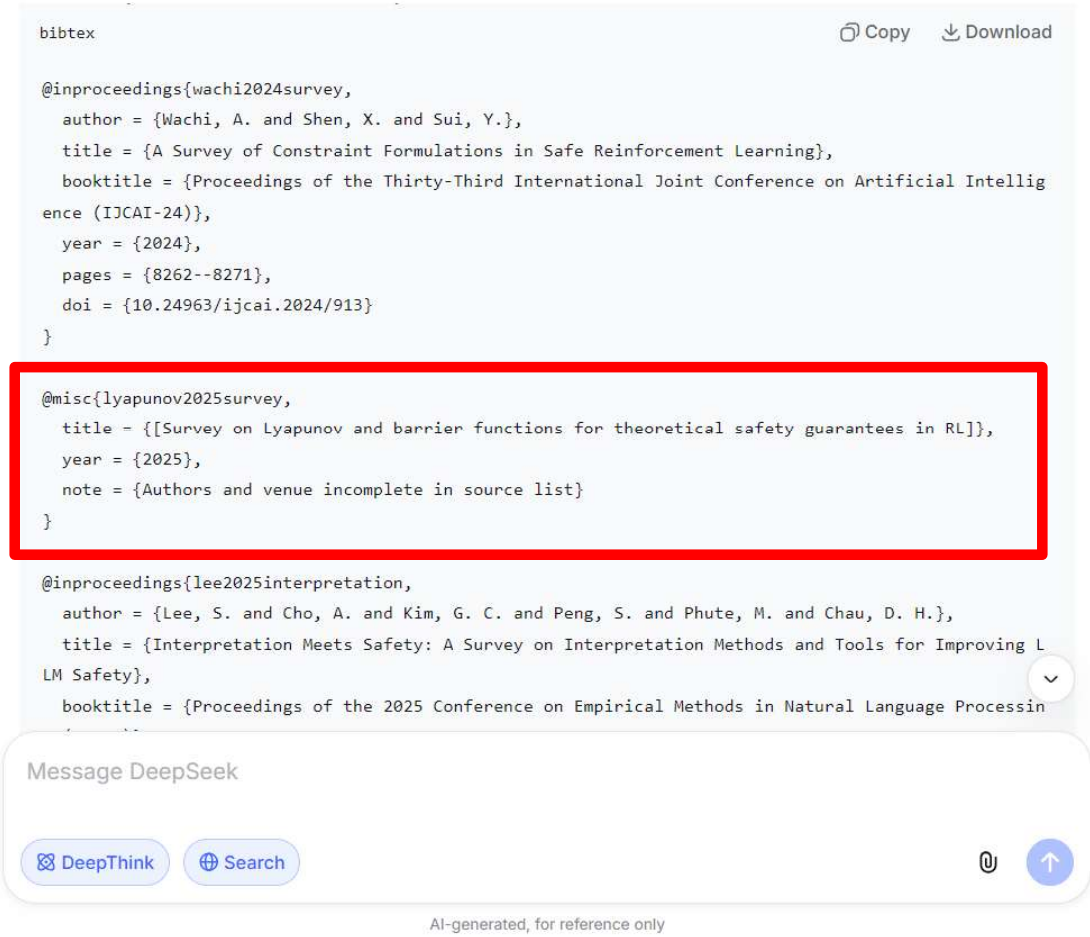
- Hallucinations are grammatically fluent and reasonable
- The content contradicts fact or cites non-existent entities
- Hallucinated contents have the same confidence level as correct ones
- Currently, no LLM system can completely eliminate hallucinations

Examples: Problematic Citations



Examples: Problematic Citations

AI Safety Related Work



Examples: Problematic Citations

```
@misc{lyapunov2025survey,
  title = {[Survey on Lyapunov and barrier functions for theoretical safety guarantees in RL]},
  year = {2025},
  note = {Authors and venue incomplete in source list}
}
```

```
@misc{adversarial2026twenty,
  title = {Adversarial Machine Learning: A 20-Year Survey of Attacks, Defenses, and Standards},
  year = {2026},
  journal = {IEEE Transactions on [Journal Name]},
  note = {Authors incomplete in source list}
}
```

Examples: Problematic Citations

```
@article{ji2025aiAlignment,
  author = {Ji, J. and Qiu, T. and Chen, B. and Zhang, B. and Lou, H. and Wang, K. and Duan, Y. and
He, Z. and Zhou, J. and Zhang, Z. and Zeng, F. and Dai, J. and Pan, X. and O'Gara, A. and Xu, H. and
Tse, B. and Fu, J. and McAleer, S. and Yang, Y. and Wang, Y. and Zhu, S.-C. and Guo, Y. and Gao, W.
and Yang, Y.},
  title = {AI Alignment: A Contemporary Survey},
  journal = {ACM Computing Surveys},
```

AI Alignment: A Comprehensive Survey

Jiaming Ji, Tianyi Qiu, Boyuan Chen, Borong Zhang, Hantao Lou, Kaile Wang, Yawen Duan, Zhonghao He, Lukas Vierling, Donghai Hong, Jiayi Zhou, Zhaowei Zhang, Fanz
Zeng, Juntao Dai, Xuehai Pan, Kwan Yee Ng, Aidan O'Gara, Hua Xu, Brian Tse, Jie Fu, Stephen McAleer, Yaodong Yang, Yizhou Wang, Song-Chun Zhu, Yike Guo, Wen Ga

AI alignment aims to make AI systems behave in line with human intentions and values. As AI systems grow more capable, so do risks from misalignment. To provide a comprehensive and up-to-date overview of the alignment field, in this survey, we delve into the core concepts, methodology, and practice of alignment. First, we identify four principles as the key objectives of AI alignment: Robustness, Interpretability, Controllability, and Ethicality (RICE). Guided by these four principles, we outline the landscape of current alignment research and decompose them into two key components: forward alignment and backward alignment. The former aims to make AI systems aligned via alignment training, while the latter aims to gain evidence about the systems' alignment and govern them appropriately to avoid exacerbating misalignment risks. On forward alignment, we discuss techniques for learning from feedback and learning under distribution shift. On backward alignment, we discuss assurance techniques and governance practices.

We also release and continually update the website (this http URL) which features tutorials, collections of papers, blog posts, and other resources.

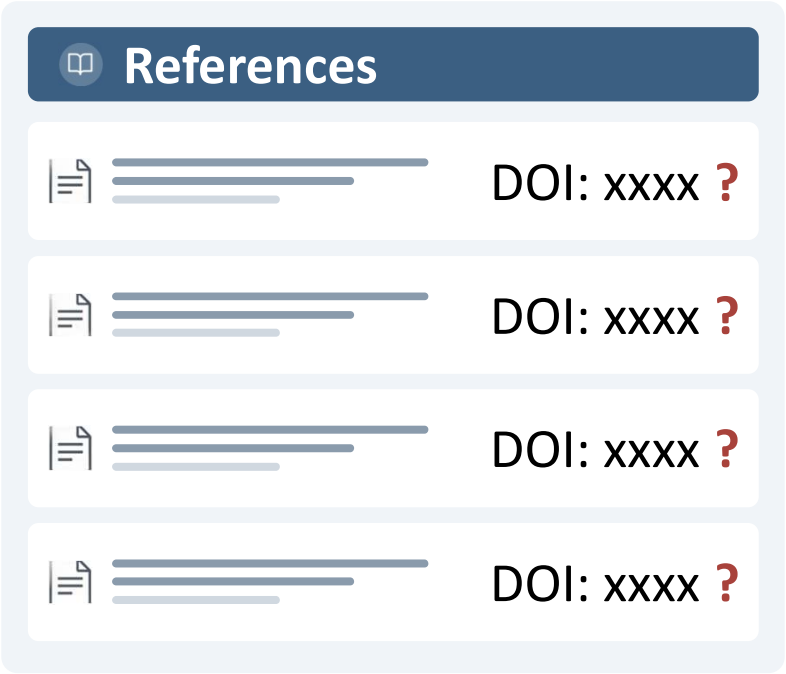
Examples: Problematic Citations

You have been warned!

Bad things happen to false citations:

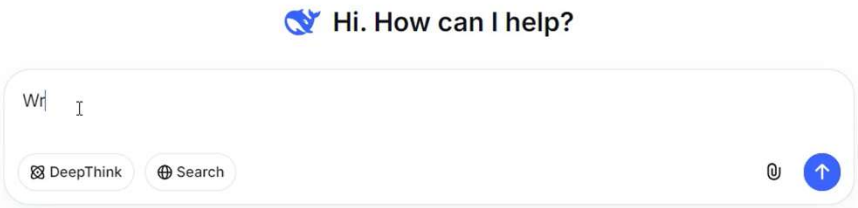
Zero-scored homework submission

Desk-reject manuscripts to top conferences

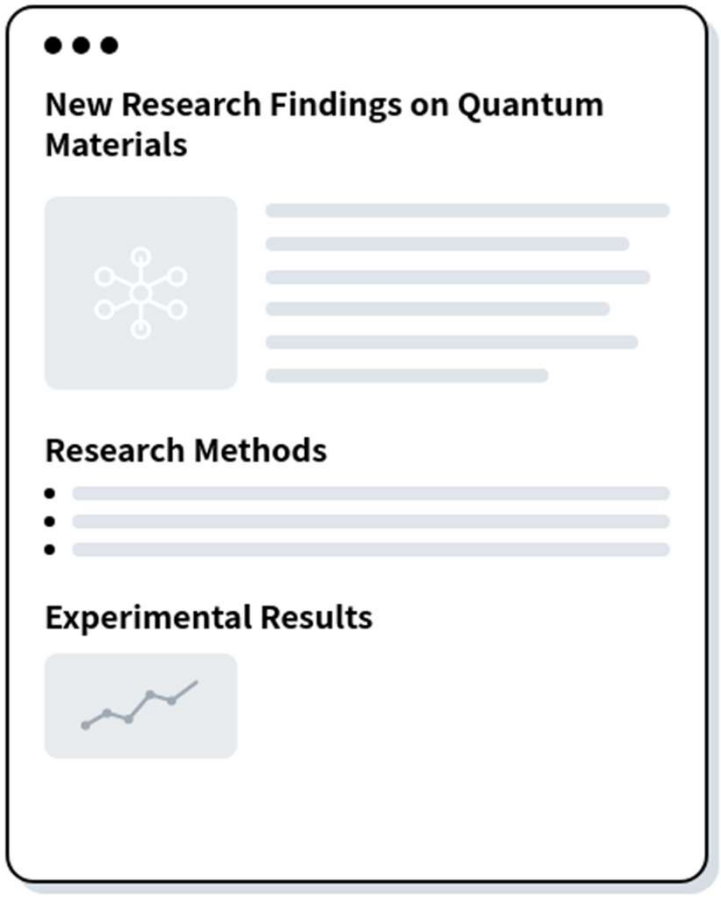


Actions: verify citations by yourself!

Examples: Experiments from Nowhere



Examples: Experiments from Nowhere



When prompting:

Generate a paragraph about the experiment …

Hallucinating experiments are likely generated

Actions: first-hand knowledge should be written by yourself

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Part 3: Responsible Use of LLM



Academic Integrity and LLM Usage



Align with the corresponding guidelines



Protect Yourself: Your Prompt May be Leaked

Academic Integrity: Wrong Actions

- Directly copying LLM-generated text into assignments or reports without clear attribution and disclosure
- LLM tools to obtain assistance or complete tasks in essays or reports exams where any external assistance is explicitly prohibited
- Using LLM-generated false data, citations, or experimental results in academic submissions

Academic Integrity: Wrong Actions

核心内容生成：禁止依赖 AI 做出应由研究者自主判断的核心学术决策，如算法核心逻辑设计、关键代码编写、模型架构创新等，不得直接采用 AI 生成的未经验证的算法或代码作为论文的研究成果。

✖ Core content generation

实验数据造假：严禁将 AI 生成的模拟数据冒充为真实实验数据，也禁止将模拟数据与真实数据混合后作为真实数据提交。如研究设计本身涉及 AI 模拟数据的使 用，应严格区分和注明。

✖ Experimental data falsification

敏感数据泄密：严禁使用敏感信息、涉密信息及未获授权的数据训练或驱动人工智能模型，确保防范数据泄露风险以及敏感数据汇聚、关联引发的泄密风险，保障数据安全。

✖ Sensitive data leakage

参考文献造假：严禁利用 AI 工具捏造、虚构参考文献，包括但不限于以下情形：利用 AI 工具凭空捏造不存在的文献条目（如虚假的作者、期刊名称、发表年份、文章标题、卷期页码等核心信息）；利用 AI 工具修改已有真实文献的关键信息（如篡改研究结论、调整数据来源、变更发表载体等），使其与文稿引用需求强行匹配；利用 AI 工具批量生成格式看似规范但无实际对应文献的引用内容，以虚假充实参考文献数量、掩盖研究基础薄弱的问题。

✖ Reference fabrication

蓄意规避查重：严禁利用 AI 工具规避学术查重，包括对他人论文内容进行 AI 改写伪装原创、生成与已有研究高度相似但无实质创新的内容等。

✖ Evasion of plagiarism detection

摘自《交叉信息研究院AI 辅助学术论文写作规范性指南》

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Part 4: LLM-Assisted Learning



Paper Reading assisted by LLM



Conceptual Explanation

Critical Thinking Prompt

Paper Reading assisted by LLM



Use LLM as a Paper Navigation Map

- Problem Background
- Key Contribution
- Main Technical Detail
- Comparison Experiments

LLM summary is a navigation map, not a shortcut to replace reading

Conceptual Explanation

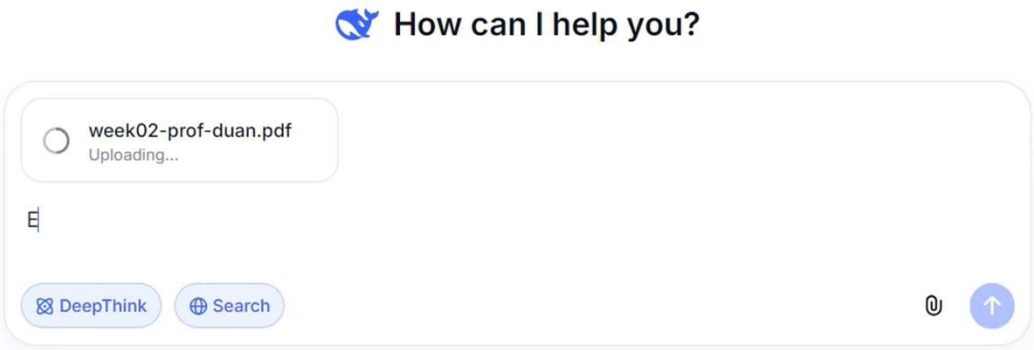
Explain the definition of the term “cache coherence” in the specific context of this paper

In the specific context of this paper, what is the difference between latency and throughput?

Transforming Passive Reading into Active Analysis



Example


How can I help you?
week02-prof-duan.pdf
Uploading...
DeepThink
Search